

Toy model of viscous fingering: two-tube approach

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Based on ongoing research with Sergey Tikhomirov and Yalchin Efendiev

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What is viscous fingering?

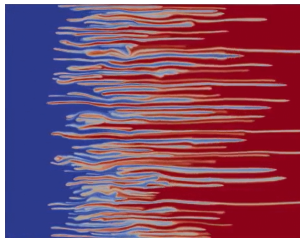
- ▶ Saffmann-Taylor instability (1958)
- ▶ **Peaceman model**: one-phase miscible displacement in porous media

$$c_t + \operatorname{div}(uc) = \varepsilon \Delta c, \quad (\text{conservation of species})$$

$$\operatorname{div}(u) = 0, \quad (\text{incompressibility condition})$$

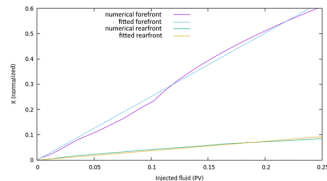
$$u = -\frac{\nabla p}{\mu(c)}. \quad (\text{Darcy's law})$$

- ▶ main source of instability (and non-linearity):
viscosity $\mu(c)$; mobility $m(c) = 1/\mu(c)$
- ▶ related phenomena: Hele-Shaw cell, gravity-driven fingers
- ▶ **Main question of interest**: rigorous bounds on velocities of mixing zone propagation



Peaceman model: numerical experiments

- ▶ linear growth of the “fastest finger”

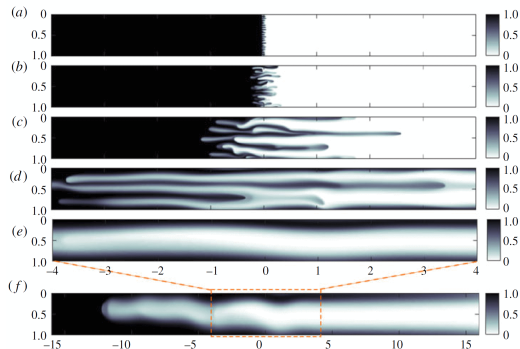


- ▶ Nijjer, Hewit, Neufield (Cambridge), 2018: $h(t)$ — mixing zone width

Three flow regimes:

1. early-time
 $h \sim C_1 \sqrt{t}$
2. intermediate-time
 $h \sim C_2 t$
3. late-time
 $h \sim C_3$

The dynamics of miscible viscous fingering from onset to shutdown



Peaceman vs Transverse Flow Equilibrium (TFE) model

(conservation of species)

$$c_t + u \cdot \nabla c = \varepsilon \cdot \Delta c,$$

(incompressibility condition)

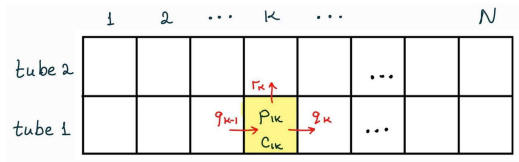
$$\operatorname{div}(u) = 0.$$

Peaceman model:	TFE model:
$u = -m(c)\nabla p$	$u = (u^1, u^2), \quad u^1 = \frac{m(c)}{\bar{m}(x,t)}$ $\bar{m}(x, t)$ is the average mobility over the transverse direction to the flow
• strong non-locality due to the presence of pressure • no rigorous result?	• weaker non-locality (depends only on vertical line, not the whole space) • Felix Otto (2006) rigorously proved the linear bounds on velocities

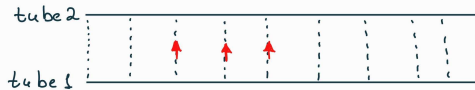
Hypothesis: models show similar behavior when $p(x, y, t) \approx p(x, t)$ (“moderate” times)

Toy model of viscous fingering: two tubes

Discrete setting



Continuous setting

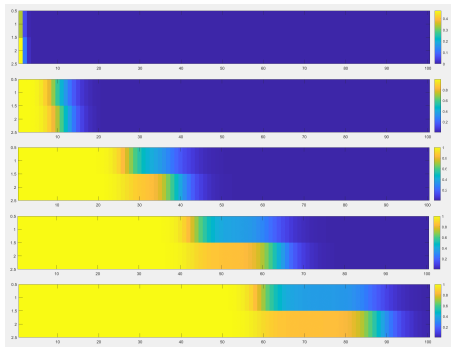


- ▶ two tubes that “speak with each other”
- ▶ initially slightly more fluid in one tube than in the other

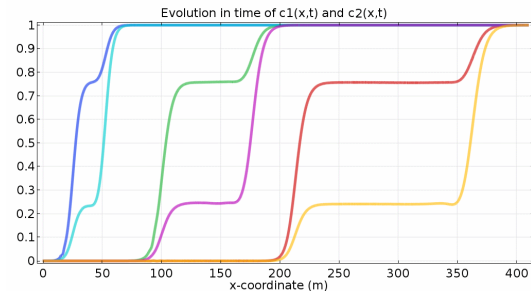
Question: how the fingers will evolve?

Toy model of viscous fingering: preview of the main idea

Discrete setting



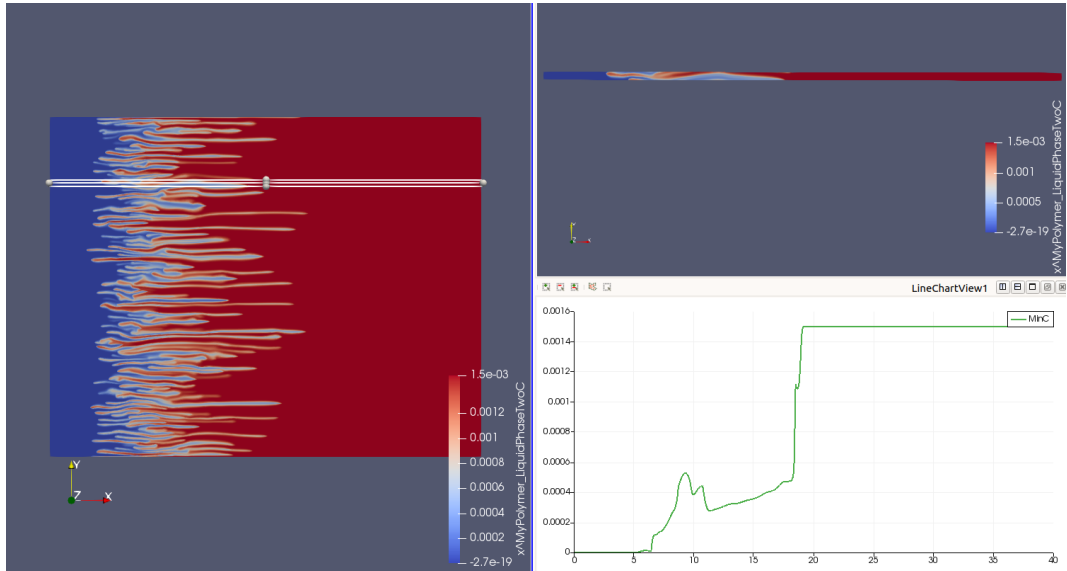
Continuous setting



Result of experiments: cascade of two travelling waves (TW)

$$(0, 0) \xrightarrow{\text{TW}_1} (c_1^*, c_2^*) \xrightarrow{\text{TW}_2} (1, 1)$$

Peaceman model: numerical experiments (show video)



Toy model of viscous fingering

14/04/2022

Notes from
Yulia Petrova

§0. Intro.

System of eqs:

I. (conservation of species):

$$c_t + \operatorname{div}(uc) = \varepsilon \Delta c$$

II. (incompressibility cond.):

$$\operatorname{div}(u) = 0$$

For velocity "u" we consider 2 models:

III a. (Darcy's law)

$$u = -m(c) \nabla p$$

III b. (TFE model)

$$u = \frac{m(c)}{\overline{m(c)}}$$

$$\text{, where } \overline{m(c)}(x,t) = \int_0^1 m(c(x,y,t)) dy$$

Note: I+II+III a is usually called Peaceman model.

Here: $c = c(x,y,t) \in [0,1]$ - concentration of species

$m(c)$ - mobility (typical $m(c) = e^{-Rc} \frac{1}{\mu_1}$)

$x \in \mathbb{R}, y \in [0,1], t \in \mathbb{R}_+$

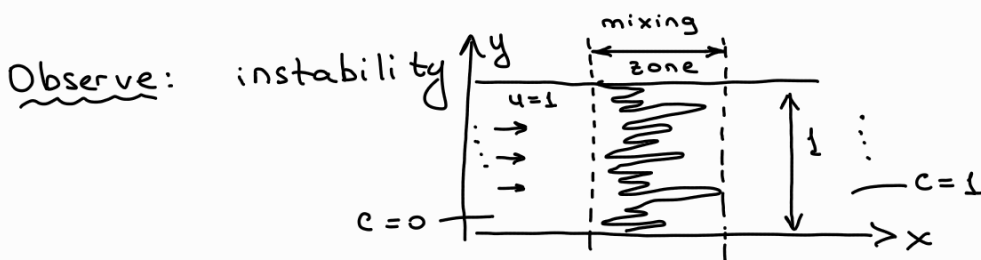
$p = p(x,y,t)$ - pressure; $u = u(x,y,t)$ - velocity



$$M = \frac{\mu_2}{\mu_1} = e^{-R}$$

ratio of viscosities

$$\varepsilon = \frac{1}{Pe} \rightarrow 0$$



Observe: instability

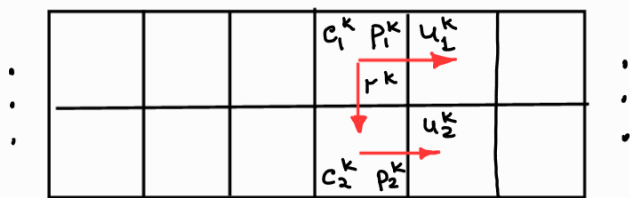
Aim: rigorously explain linear growth of the mixing zone

Two-tubes approach.

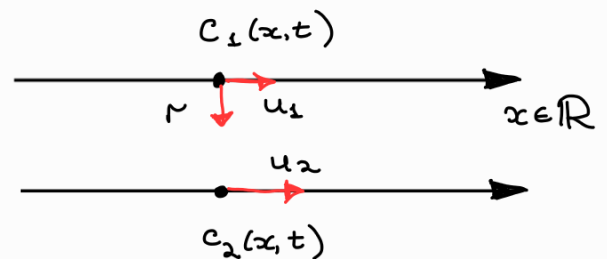
§1. Problem statement. Derivation of the model.

Discrete (in x)

$k \in \mathbb{Z}$



Continuous (in x)



Unknowns:

- $c_{1,2}^k(t)$ - concentrations in 1,2 tube $\longleftrightarrow c_{1,2}(x,t)$
- $u_{1,2}^k(t)$ - velocity in 1,2 tube $\longleftrightarrow u_{1,2}(x,t)$
- $r^k(t)$ - velocity between tubes $\longleftrightarrow r(x,t)$
- $p_{1,2}^k(t)$ - pressure in 1,2 tube $\longleftrightarrow p_{1,2}(x,t)$

$$\text{I. } \frac{d}{dt} c_1^k = u_1^{k-1} c_1^{k-1} - u_1^k c_1^k - r^k c_1^k$$

$$\frac{d}{dt} c_2^k = u_2^{k-1} c_2^{k-1} - u_2^k c_2^k + r^k c_2^k$$

width
of cells
is $\delta \rightarrow 0$

$$\text{I. } \partial_t c_1 = -\partial_x (c_1 u_1) + c_1 \partial_x u_1$$

$$\partial_t c_2 = -\partial_x (c_2 u_2) - c_1 \partial_x u_1$$

$$\text{II. } \begin{cases} u_1^k - u_1^{k-1} + r^k = 0 \\ u_2^k - u_2^{k-1} - r^k = 0 \end{cases} \Rightarrow \begin{cases} u_1^k + u_2^k = u_1^{k-1} + u_2^{k-1} = 2 \\ r^k = u_1^{k-1} - u_1^k \end{cases}$$

$\Rightarrow$
after
change
of
variables

$$\text{II} + \text{III} b: u_1 = \frac{2m(c_1)}{m(c_1) + m(c_2)}$$

$$u_2 = 2 - u_1$$

$$\text{III} b. u_1(c_1, c_2) = \frac{2m(c_1)}{m(c_1) + m(c_2)} = 2 - u_2$$

Taking into account that fluid may flow not only from 1st tube to 2nd, but vice versa as well, we get:

$$\partial_t c_1 = -\partial_x (c_1 u_1) - (-1)^{1/2} c_{1,2} \partial_x u_1 + \varepsilon \partial_{xx} c_1$$

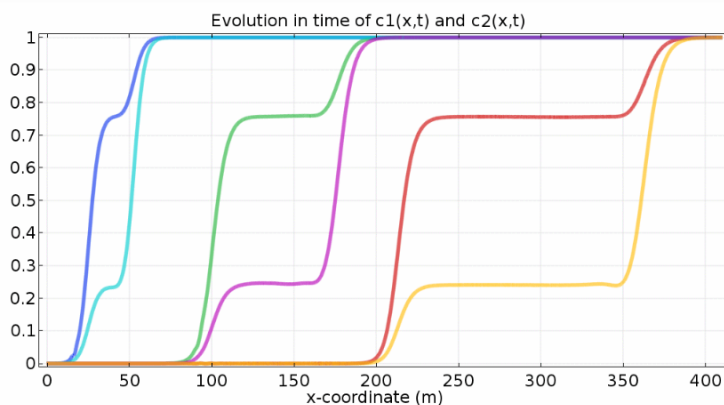
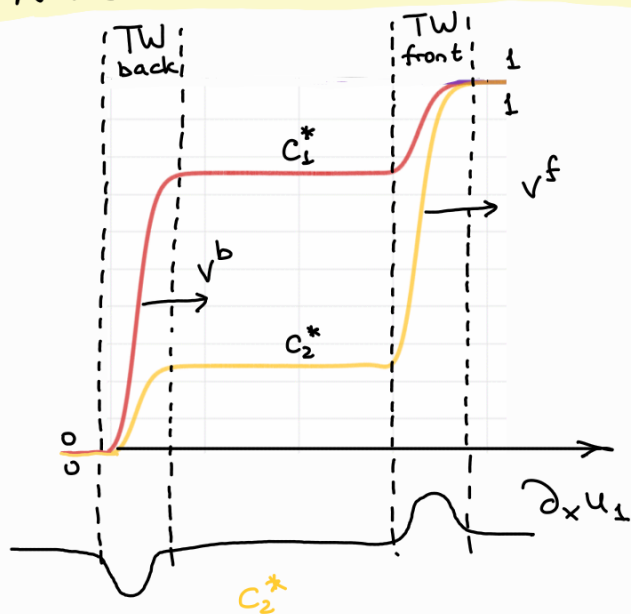
$$\partial_t c_2 = -\partial_x (c_2 u_2) + (-1)^{1/2} c_{1,2} \partial_x u_1 + \varepsilon \partial_{xx} c_2$$

2

For simplicity consider TFE: $u_1 = \frac{2m(c_1)}{m(c_1) + m(c_2)} = 2 - u_2$

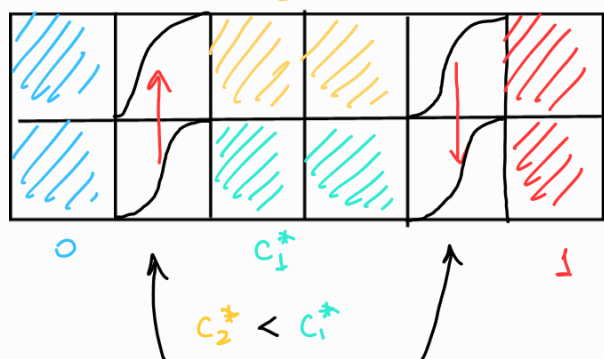
NB: consider Darcy's law here would be interesting and could even be easier because the system is more physically meaningful.

§2. Numerical solution: cascade of travelling waves (TW)



3 zones are formed:

- in both tubes concentrations 0
- in both tubes concentrations 1
- $\exists (c_1^*, c_2^*)$:
in 1st tube concentration is c_1^*
in 2nd tube concentration is c_2^*



And 2 transition zones between $(0,0)$ and (c_1^*, c_2^*) ; (c_1^*, c_2^*) and $(1,1)$

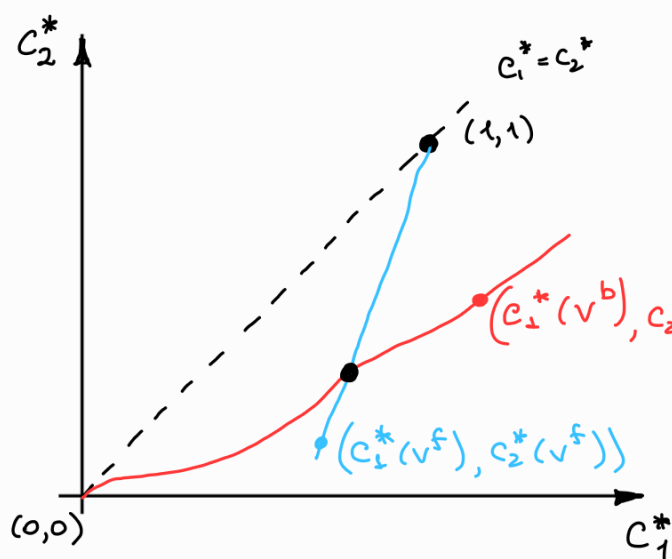
TW back : travelling wave with velocity $v^b \in (0,1)$
between $(0,0)$ and (c_1^*, c_2^*)

TW front : travelling wave with velocity $v^f \in (1,+\infty)$
between (c_1^*, c_2^*) and $(1,1)$

So we would like to prove the following structure:

$$\exists (c_1^*, c_2^*) : (0,0) \xrightarrow[\text{back}]{\text{TW}} (c_1^*, c_2^*) \xrightarrow[\text{front}]{\text{TW}} (1,1)$$

§3. General scheme of proof of the existence of cascade of travelling waves.



Hypothesis :

A) $\forall v^b \in (0,1) \exists! (c_1^*(v^b), c_2^*(v^b))$
there exists a travelling wave

$$(0,0) \xrightarrow{\text{TW}} (c_1^*(v^b), c_2^*(v^b))$$

Corresponds to **red point**

All such points $(c_1^*(v^b), c_2^*(v^b))$
form a **red curve**

B) $\forall v^f \in (1,+\infty) \exists! (c_1^*(v^f), c_2^*(v^f))$
there exists a travelling wave

$$(c_1^*(v^f), c_2^*(v^f)) \xrightarrow{\text{TW}} (1,1)$$

Corresponds to **blue point**.

All such points $(c_1^*(v^f), c_2^*(v^f))$
form a **blue curve**.

If **red** and **blue** curves intersect in 1 point -
- this will give us intermediate concentrations
 (c_1^*, c_2^*) and velocities of travelling waves v^b and v^f .

No proofs yet...

P.S. If system (2) was a system of conservation laws, these two curves would be just Hugoniot loci defined by Rankine-Hugoniot condition.

§4. Travelling wave dynamical system

Let's look for solution being a travelling wave

$$c_{1,2} = c_{1,2} \left(\frac{x - v^b t}{\varepsilon} \right), \quad v^b - \text{velocity, such that}$$

$$c_{1,2}(-\infty) = 0 \quad \text{and} \quad c_{1,2}(+\infty) = c_{1,2}^*.$$

After some modifications we get:

$$\begin{cases} \dot{c}_1 = g_1 \\ \dot{g}_1 = (u(c_1, c_2) - v^b) g_1 \\ \dot{c}_2 = (u(c_1, c_2) - v^b) c_1 + (2 - u(c_1, c_2) - v^b) c_2 - g_1 \end{cases}$$

3

Fixed points:

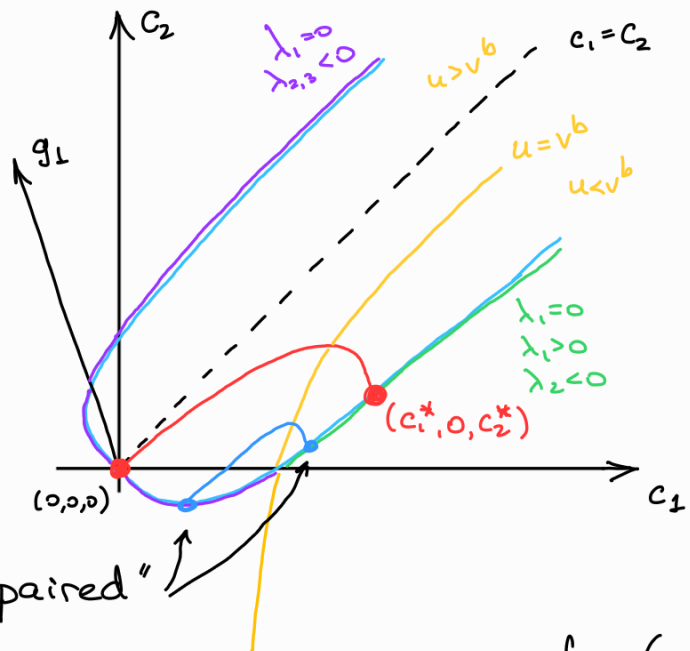
$$\begin{cases} g_1 = 0 \\ (u - v^b) c_1 + (2 - u - v^b) c_2 = 0 \end{cases}$$

We are looking for

$$(c_1^*(v^b), c_2^*(v^b))$$

such that there exists a trajectory from

$$(0, 0, 0) \rightarrow (c_1^*, 0, c_2^*)$$



Hypothesis: points are "paired"

Ideas how to prove the existence of (c_1^*, c_2^*) and understand the dependence $c_1^*(v^b)$ and $c_2^*(v^b)$?

Construct Lyapunov function?

§5. Open questions and directions of research:

- 1) Existence and uniqueness of ^{global} solutions for TFE model (literature review, Yudovich theory works?)
- 2) Study the limit as $\varepsilon \rightarrow 0$ (spontaneous stochasticity?)
- 3) Careful numerical modelling of TFE model and comparison "Peaceman" and "TFE" - are they close in some sense?
- 4) 2-tube model with Darcy's law - numerically we observe similar cascade of 2 travelling waves - prove